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MH1200 Linear Algebra I

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1. Quotes

"But we can do better"

PROF ANDREW SAID THE SIR KEVIN LINE 🤯

2. Finding the Inverse

We note that Gauss-Jordan finds, by proxy of the characterizations of invertibility, elementary matrices that turn a matrix into RREF.

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 7 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 7 & 0 \\ 0 & 0 & 1 & -7 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -7 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -7 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ -1 & 3 & -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -7 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & -2 & 0 \\ 0 & -3 & 1 & 0 \\ -1 & 3 & -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & -2 & 0 \\ 0 & -3 & 1 & 0 \\ \frac{1}{7} & -\frac{3}{7} & \frac{1}{7} & -\frac{1}{7} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} -\frac{1}{7} & \frac{24}{7} & -\frac{8}{7} & \frac{8}{7} \\ 0 & 7 & -2 & 0 \\ 0 & -3 & 1 & 0 \\ \frac{1}{7} & -\frac{3}{7} & \frac{1}{7} & -\frac{1}{7} \end{pmatrix}$$

Note to self: This is easier than the adjoint matrix method.

3. Determinants

3.1. Induction

Theorem | Induction

Theorem 1

$$P(0)$$

$$\forall n \in \mathbb{N}, P(n) \to P(n+1)$$

$$\therefore \forall n \in \mathbb{N}, P(n)$$

Example Sum of Squares

Example 2

We want to show:

$$P(n) : 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

For all $n \in \mathbb{N}$.

First, we show:

$$P(0) : 0 = 0 \cdot \frac{0+1}{2}$$

Indeed, $0 \cdot \frac{0+1}{2} = 0 \cdot \frac{1}{2} = 0$, so $P(0)$, as needed.

Next, we show that $P(n) \to P(n+1)$.

Suppose $P(n)$. We want to show:

$$P(n+1) : 1 + 2 + 3 + \cdots + n + (n+1) = (n+1)\frac{n+2}{2}$$

Notice that:

$$\begin{aligned} 1 + 2 + 3 + \cdots + n + (n+1) &= (1 + 2 + 3 + \cdots + n) + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{1}{2}(n(n+1) + 2(n+1)) \\ &= \frac{1}{2}((n+2)(n+1)) \\ &= \frac{(n+1)(n+2)}{2} \end{aligned}$$

Thus, $P(n) \to P(n+1)$. Since $P(0)$ and $\forall n \in \mathbb{N}, P(n) \to P(n+1)$. Thus, by the principle of mathematical induction,

$$\forall n \in \mathbb{N}, P(n)$$

As needed. ■

Base Case The first thing you prove in mathematical induction to start the "fall of the chain of dominoes"

Inductive Step Show that if one domino falls, the next one falls.

3.2. Definition of the Determinant of a matrix

Definition Minor Matrix

Definition 3

The $\text{minor}(M, i, j)$ minor matrix of M is the result of removing the ith row and jth column from M.

Definition Determinant

Definition 4

The **determinant** of an $n \times n$ **square matrix**, denoted $\det(A)$, is defined as:

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+1} a_{1i} \det(\text{minor}(A, 1, i))$$

Where the determinant of a 1×1 matrix is defined as:

$$\det(A) = a_{11}$$

Definition Cofactor

Definition 5

The C_{ij} cofactor of M is given by:

$$C_{ij} := (-1)^{i+j} \det(\text{minor}(M, i, j))$$

So, the determinant can be equivalently computed as:

$$\det(A) = \sum_{i=1}^{n} a_{1i} C_{1i}$$

Theorem Properties of the Determinant

Theorem 6

1. A matrix A is invertible iff $\det(A) \neq 0$.
2. $\det(AB) = \det(A) \det(B)$